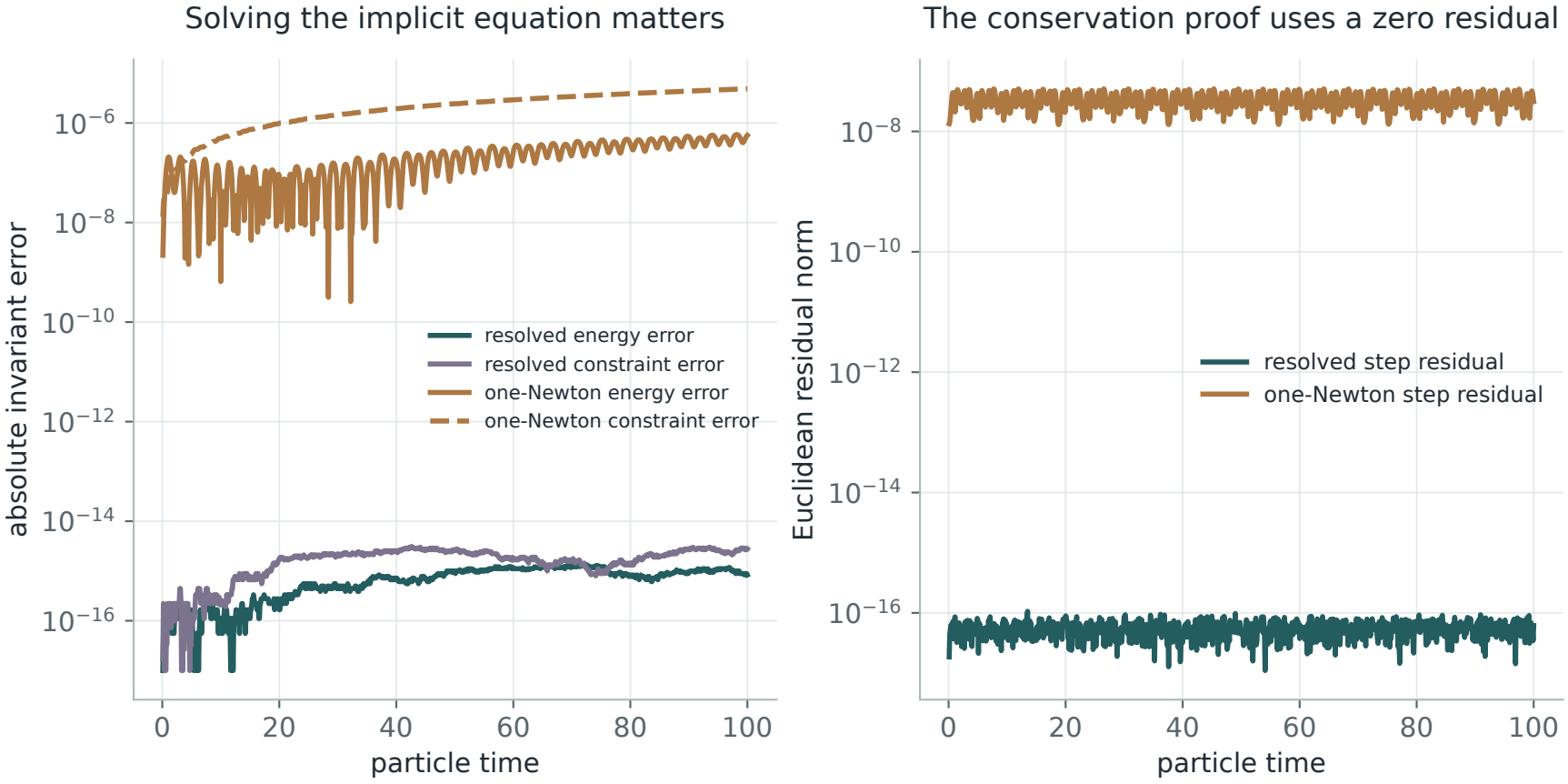
Solving the implicit equation matters
absolute invariant error
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
resolved energy error
resolved constraint error
one-Newton energy error
one-Newton constraint error
0
20
40
60
80
100
particle time
The conservation proof uses a zero residual
Euclidean residual norm
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
resolved step residual
one-Newton step residual
0
20
40
60
80
100
particle time